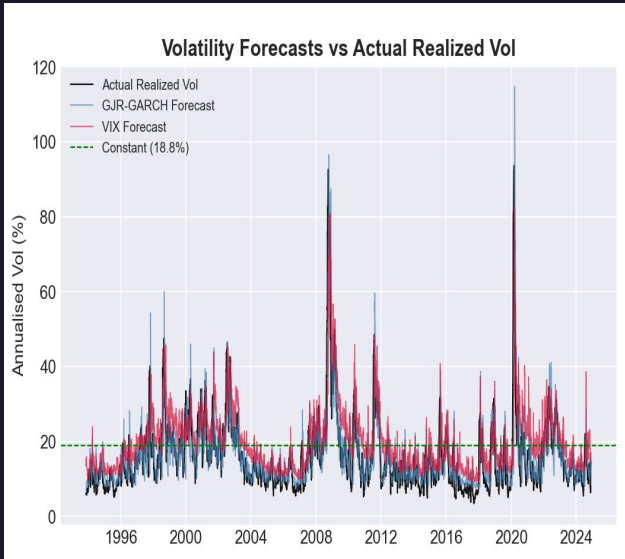
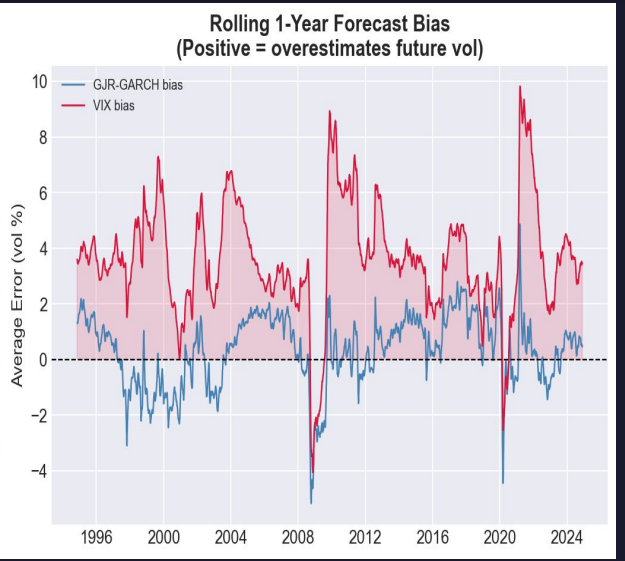
Does the Market Know Its Own Volatility?

Nathan Fleck · April 2026

GJR-GARCH outforecasts VIX — despite VIX aggregating the entire options market





BEST FORECASTER

GJR-GARCH

RMSE = 7.67% · Bias = +0.35%

MARKET IMPLIED (VIX)

Biased +3.66%

RMSE = 7.90% — overestimates vol

CONSTANT VOL

Worst forecast

RMSE = 10.39% · Bias = +2.74%

KEY INSIGHT

VIX is not a volatility forecast — it is the price of volatility insurance. Its persistent upward bias (+3.66 vol pts) reflects the variance risk premium, not forecasting error.

Data: SPY daily returns 1993–2024 (7,838 days) · Models: GJR-GARCH(1,1) vs. CBOE VIX vs. Constant Historical Vol · Metric: 21-day realized vol